

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To understand the constraint satisfaction problem in problem formulation	
Experiment No: 13	Date:	Enrolment No: 92410133034

Aim: To understand the constraint satisfaction problem in problem formulation

IDE: Google Colab

Theory:

The Constraint Satisfaction Problem (CSP) is a fundamental concept in computer science and artificial intelligence, playing a crucial role in solving a wide range of real-world problems. At its core, CSP involves finding a solution to a problem where the solution must satisfy a set of constraints that define the problem's requirements and limitations. These constraints restrict the possible values that variables can take, leading to a search for a combination of values that satisfies all constraints simultaneously.

CSPs can be formulated in various domains, including scheduling, planning, resource allocation, configuration, and design. Examples of CSP applications include job scheduling, timetabling, Sudoku puzzles, map coloring, and the traveling salesman problem. In each case, the problem consists of a set of variables, each with a domain of possible values, and a set of constraints that specify allowable combinations of variable values.

The basic components of a CSP include:

1. **Variables:** These represent the entities whose values need to be determined. Each variable has a domain, which is a set of possible values it can take.
2. **Domains:** A domain is the set of values that a variable can take. Domains can be finite or infinite, discrete or continuous, depending on the problem domain.
3. **Constraints:** Constraints define the relationships among variables and restrict the combinations of values they can take. Constraints can be unary (applying to a single variable), binary (between two variables), or higher-order (involving three or more variables).

The goal in solving a CSP is to find an assignment of values to variables such that all constraints are satisfied. This involves searching through the space of possible assignments to find a solution or determine that no solution exists. The search process typically involves backtracking, constraint propagation, and variable ordering heuristics to efficiently explore the search space.

One common approach to solving CSPs is backtracking search, which systematically explores the space of possible assignments by making choices and backtracking when a dead-end is reached. Backtracking search is guided by variable and value ordering heuristics to prioritize the search for promising solutions and avoid exploring unpromising regions of the search space.

Another technique used in CSP solving is constraint propagation, which involves enforcing constraints locally to reduce the domain of variables and prune the search space. Constraint propagation techniques include arc consistency, domain reduction, and constraint propagation algorithms such as AC-3 and AC-4.

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In addition to backtracking search and constraint propagation, other methods for solving CSPs include local search algorithms, genetic algorithms, and constraint satisfaction neural networks. Each approach has its strengths and weaknesses, making them suitable for different types of CSPs and problem domains.

Despite the challenges involved in solving CSPs, they offer a powerful framework for modeling and solving complex combinatorial optimization problems. By formulating real-world problems as CSPs and applying appropriate solving techniques, researchers and practitioners can tackle a wide range of practical problems efficiently and effectively.

Methodology:

1. Mention the constraints for sudoku solution
 - Each number (1–9) appears **only once in each row**
 - Each number (1–9) appears **only once in each column**
 - Each number (1–9) appears **only once in each 3×3 subgrid**
 - All cells must contain **values from 1 to 9**
 - No repetition of numbers is allowed

2. Solve the sudoku using the CSP approach
 - Define Sudoku as a Constraint Satisfaction Problem (CSP)
 - Each cell is treated as a variable
 - Domain of each variable is numbers from 1 to 9
 - Apply constraints (row, column, 3×3 grid rules)
 - Use backtracking algorithm to assign values
 - Check constraints before assigning each value
 - If a conflict occurs, backtrack and try another value
 - Continue until the Sudoku is completely solved

Program (Code):

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[14]
✓ 2s
board[row][col] = num

if solve_sudoku(board):
    return True

board[row][col] = 0 # backtrack
return False

return True

# Example Sudoku (0 means empty)
board = [
    [5,3,0,0,7,0,0,0,0],
    [6,0,0,1,9,5,0,0,0],
    [0,9,8,0,0,0,0,6,0],
    [8,0,0,0,6,0,0,0,3],
    [4,0,0,8,0,3,0,0,1],
    [7,0,0,0,2,0,0,0,6],
    [0,6,0,0,0,0,2,8,0],
    [0,0,0,4,1,9,0,0,5],
    [0,0,0,0,8,0,0,7,9]
]

# Solve
solve_sudoku(board)

print("Solved Sudoku:")
print_board(board)

```

```

... print_board(board)

... Solved Sudoku:
[5, 3, 4, 6, 7, 8, 9, 1, 2]
[6, 7, 2, 1, 9, 5, 3, 4, 8]
[1, 9, 8, 3, 4, 2, 5, 6, 7]
[8, 5, 9, 7, 6, 1, 4, 2, 3]
[4, 2, 6, 8, 5, 3, 7, 9, 1]
[7, 1, 3, 9, 2, 4, 8, 5, 6]
[9, 6, 1, 5, 3, 7, 2, 8, 4]
[2, 8, 7, 4, 1, 9, 6, 3, 5]
[3, 4, 5, 2, 8, 6, 1, 7, 9]

```

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Observation and Result Analysis:

The Sudoku problem was successfully solved using the Constraint Satisfaction Problem (CSP) approach with a backtracking algorithm. Each cell was treated as a variable with a domain of values from 1 to 9, and constraints were applied to ensure no repetition in rows, columns, and 3×3 grids. The algorithm systematically tried possible values and backtracked whenever a constraint was violated. The final output satisfied all Sudoku constraints, proving the correctness of the solution. This approach is efficient for solving such problems, although it may take more time for complex puzzles due to its recursive nature.

Post Lab Exercise:

Write the code and output for N-Queens problem. Take the value of N from the user.

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[15]
✓ 8s # N-Queens Problem using Backtracking

def is_safe(board, row, col, n):
    # Check column
    for i in range(row):
        if board[i][col] == 1:
            return False

    # Check left diagonal
    i, j = row-1, col-1
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1

    # Check right diagonal
    i, j = row-1, col+1
    while i >= 0 and j < n:
        if board[i][j] == 1:
            return False
        i -= 1
        j += 1

    return True

def solve_nqueens(board, row, n):
    if row == n:
        return True

    for col in range(n):
        if is_safe(board, row, col, n):
            board[row][col] = 1

```

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[15]

✓ 8s

▶

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        if solve_nqueens(board, row+1, n):
            return True

        board[row][col] = 0 # backtrack

    return False

# Take input
n = int(input("Enter value of N: "))

# Create board
board = [[0]*n for _ in range(n)]

# Solve
if solve_nqueens(board, 0, n):
    print("Solution:")
    for row in board:
        print(row)
else:
    print("No solution exists")

```

▼

```

... Enter value of N: 4
Solution:
[0, 1, 0, 0]
[0, 0, 0, 1]
[1, 0, 0, 0]
[0, 0, 1, 0]

```